

Introduction

Financial markets are among the most complex and dynamic systems studied in data science and quantitative finance. Stock prices are influenced simultaneously by macroeconomic conditions, investor psychology, exchange-rate movements, monetary policy decisions, global market sentiment, political uncertainty, and unexpected economic shocks. Unlike deterministic systems, financial markets behave stochastically, meaning that uncertainty and unpredictability are fundamental characteristics of price behavior.

Because relationships between financial variables continuously evolve over time, predicting stock returns remains one of the most challenging applications of statistical analysis and machine learning. Models that perform well under certain market conditions may lose effectiveness when market dynamics change, making financial forecasting an inherently difficult task.

The Turkish stock market provides a particularly interesting environment for financial analysis because emerging markets are generally more sensitive to inflation, exchange-rate volatility, monetary policy decisions, and political developments than developed markets. As a result, understanding the behavior of Turkish equities provides valuable insight into how financial markets respond to changing economic conditions.

The primary objective of this project is to investigate whether statistical analysis and machine learning techniques can identify meaningful relationships between major Turkish stocks and broader market conditions represented by the BIST100 index. Rather than focusing exclusively on stock prices, the project aims to understand market dependency, volatility behavior, downside risk, predictive limitations, and financial uncertainty through a combination of exploratory data analysis, statistical inference, regression analysis, and machine learning classification techniques.

An additional objective is to examine the distinction between correlation and predictability. Although financial variables may move together and exhibit strong statistical relationships, these relationships do not necessarily imply that future movements can be accurately predicted.

Research Questions

This project investigates four primary research questions:

1. Do Turkish large-cap stocks systematically move together with the BIST100 market index?
2. How do volatility, rolling beta values, and drawdowns evolve under changing market conditions?

3. Are stock returns statistically different from overall market returns?
4. Can machine learning algorithms predict the next-day direction of stock returns using historical financial indicators?

The ultimate goal is not simply to maximize prediction accuracy but to better understand the structure, behavior, and limitations of Turkish financial markets.

Dataset and Data Collection

Historical daily financial data was collected using the Yahoo Finance API through the Python `yfinance` library.

The dataset covers the period between 2018 and 2025, a time horizon that includes several economically significant events, including post-pandemic recovery periods, inflationary episodes, exchange-rate instability, monetary policy adjustments, and increased market volatility.

The selected assets are:

- THYAO.IS
- GARAN.IS
- ASELS.IS
- KCHOL.IS
- TUPRS.IS
- BIMAS.IS
- BIST100 Index (^XU100)
- S&P500 Index (^GSPC)
- USD/TRY Exchange Rate (USDTRY=X)

For each asset, the following variables were collected:

- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume

Adjusted closing prices were used for return calculations because they account for stock splits and dividend distributions, providing a more accurate representation of investor returns.

The selected stocks represent different sectors of the Turkish economy, including aviation, banking, defense, energy, retail, and diversified industrial holdings. This sectoral diversity allows the analysis to capture a broad range of market behaviors and economic influences.

Data Preprocessing

Several preprocessing procedures were performed before conducting the statistical and machine learning analyses.

First, all datasets were aligned according to trading dates to ensure consistency across assets. Missing observations were handled through forward-filling techniques where appropriate. Infinite values and invalid observations were removed to improve data quality.

Daily percentage returns were calculated instead of raw price levels. Using returns rather than prices provides several advantages. Returns normalize performance across assets with different price scales, improve comparability, and reduce scale dependency in statistical models.

Additional feature engineering procedures were also performed. These include:

- Lagged returns
- Rolling averages
- Rolling volatility measures
- Momentum indicators
- Market return variables

These engineered variables were designed to improve both statistical interpretation and predictive performance during the machine learning stage.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to investigate return behavior, market dependency, volatility persistence, correlation structures, downside risk, and portfolio sensitivity. The objective of this stage was not only to visualize the data but also to identify meaningful patterns that could later support statistical inference and machine learning applications.

Return Distribution Analysis

The selected Turkish stocks exhibited substantial differences in return characteristics, volatility behavior, and downside risk exposure.

ASELS generated one of the strongest long-term return profiles among the selected assets. The company benefited from increasing investor interest in the Turkish defense sector and demonstrated strong growth potential. However, the stock also exhibited elevated volatility and several extreme return observations, indicating that higher returns were accompanied by greater uncertainty.

THYAO demonstrated strong return potential but also displayed the highest volatility among the selected stocks. As an airline company, THYAO is highly sensitive to tourism activity, fuel prices, global economic conditions, and geopolitical developments. Consequently, the stock experienced both substantial gains and significant losses throughout the observation period.

BIMAS exhibited a more defensive profile compared to the other selected stocks. As a major retail company operating within the consumer staples sector, demand for its products remains relatively stable during periods of economic uncertainty. This characteristic contributed to lower volatility and fewer extreme return observations.

GARAN displayed return behavior closely linked to macroeconomic conditions. Since banking institutions are strongly affected by interest rates, inflation expectations, and monetary policy decisions, GARAN's returns frequently reflected broader market developments.

TUPRS generated attractive returns during favorable energy market conditions. However, fluctuations in oil prices and refining margins contributed to periods of elevated volatility.

KCHOL demonstrated a relatively balanced return profile due to its diversified business structure. Because the company operates across multiple industries, its performance reflects developments in several sectors of the Turkish economy simultaneously.

The return distributions were found to be non-normal, asymmetric, and heavy-tailed. This finding suggests that extreme market events occur more frequently than predicted by traditional normal distribution assumptions and highlights the importance of robust risk management techniques.

Cumulative Return Analysis

Cumulative return analysis was conducted to evaluate long-term investment performance and compare the selected stocks with the BIST100 benchmark.

The results revealed substantial differences in long-term wealth accumulation among the selected assets. THYAO and TUPRS achieved some of the strongest cumulative returns throughout the observation period, indicating that investors who accepted higher levels of risk were rewarded

with stronger long-term growth. However, these assets also experienced larger fluctuations and periods of significant drawdowns.

ASELS demonstrated strong cumulative growth driven by increasing interest in the Turkish defense industry and favorable sector-specific developments. Although the stock generated impressive returns during several periods, it also exhibited greater volatility compared to more defensive investments.

BIMAS followed a smoother and more stable growth trajectory. While its cumulative return was not always the highest, the stock maintained relatively consistent performance and experienced fewer dramatic fluctuations. This behavior reflects the defensive characteristics of the retail sector.

GARAN and KCHOL generally followed patterns similar to broader market movements. Their cumulative returns were heavily influenced by macroeconomic developments, monetary policy changes, and investor sentiment within the Turkish economy.

The comparison with BIST100 suggests that several selected stocks were capable of outperforming the benchmark over the long term. Nevertheless, systematic market risk remained the dominant driver of overall performance.

Correlation Analysis

A correlation heatmap was constructed to examine relationships among Turkish stocks, portfolio returns, BIST100 returns, USD/TRY exchange rates, and the S&P500 index.

The results revealed strong positive correlations between most selected stocks and the BIST100 benchmark. This finding indicates that Turkish large-cap equities are heavily influenced by common market factors and tend to react similarly to changes in macroeconomic conditions.

GARAN exhibited one of the strongest relationships with BIST100. As a major banking institution, the company is highly sensitive to monetary policy decisions, inflation expectations, and economic growth conditions. Consequently, its stock performance closely tracked broader market movements.

THYAO also demonstrated strong correlation with market returns. Tourism demand, fuel costs, and investor expectations regarding economic growth influence both airline performance and overall market behavior.

ASELS maintained a positive correlation with the market while also exhibiting periods of independent movement driven by defense-related developments. This suggests that both market-wide and company-specific factors contribute to its performance.

BIMAS displayed comparatively lower correlation values than most other assets. This finding suggests potential diversification benefits because consumer staples companies often respond differently to economic fluctuations than financial, industrial, or energy firms.

One of the most important findings of this analysis is that strong correlation does not necessarily imply predictability. Assets may move together under similar market conditions while still remaining difficult to forecast accurately.

Volatility Analysis

Rolling volatility analysis was performed to investigate how financial risk evolves over time.

The results revealed clear evidence of volatility clustering, a phenomenon commonly observed in financial markets where periods of elevated volatility are followed by additional periods of heightened uncertainty. This behavior suggests that market shocks tend to persist rather than disappear immediately.

THYAO experienced some of the largest volatility spikes among the selected assets. These fluctuations were particularly visible during periods of economic uncertainty and global market instability. The findings indicate that airline stocks are highly sensitive to rapidly changing market conditions.

GARAN also displayed substantial volatility increases during unstable economic periods. Since banking institutions are directly affected by changes in monetary policy, inflation expectations, and interest rate decisions, investor reactions often amplify price movements.

ASELS exhibited moderate to high volatility throughout the sample period. While defense-sector developments supported strong growth opportunities, they also contributed to periods of elevated uncertainty.

In contrast, BIMAS maintained relatively stable volatility levels. The consistency of consumer demand helped reduce fluctuations compared to more cyclical sectors.

Overall, the volatility analysis demonstrates that financial risk is dynamic rather than constant and that different sectors respond differently to periods of market stress.

Rolling Beta Analysis

Rolling beta analysis was conducted to evaluate how market sensitivity changes through time.

The results revealed that beta values fluctuate considerably rather than remaining constant. During periods of financial stability, stock sensitivity to market movements remained relatively moderate. However, during periods of increased uncertainty, beta values generally increased.

This finding indicates that systematic market risk becomes more dominant during stressful market environments. Investors appear to react more strongly to market-wide information during uncertain periods, causing individual stock movements to become increasingly synchronized with overall market behavior.

The results further support the conclusion that market-wide dynamics play a critical role in explaining Turkish stock performance.

Drawdown Analysis

Maximum drawdown analysis was performed to evaluate downside risk and identify periods of substantial temporary losses.

THYAO experienced the largest drawdowns among the selected stocks, highlighting the substantial risks associated with high-growth opportunities. Although the stock generated strong long-term returns, investors were required to tolerate significant temporary losses.

TUPRS also recorded notable drawdowns due to its sensitivity to global energy markets and fluctuations in oil prices.

ASELS experienced several drawdown periods associated with changes in investor expectations and sector-specific developments. However, the stock demonstrated the ability to recover strongly following periods of weakness.

BIMAS displayed considerably lower drawdowns than most other assets. This finding further supports its classification as a relatively defensive investment.

GARAN and KCHOL occupied an intermediate position, experiencing moderate drawdowns while remaining highly influenced by broader market conditions.

The drawdown analysis highlights the trade-off between return potential and downside risk. Assets generating the strongest long-term returns were generally associated with larger temporary losses.

Volume and Volatility Relationship

The relationship between trading activity and volatility was examined using scatter plots comparing average trading volume and absolute return magnitude.

The analysis revealed a positive relationship between trading volume and volatility. Higher trading activity was generally associated with larger price movements.

This finding is consistent with financial theory, which suggests that increased trading activity often reflects heightened information flow, changing investor expectations, and elevated uncertainty.

Periods characterized by abnormal trading volume frequently coincided with significant market events, supporting the hypothesis that trading activity serves as an important indicator of market sentiment and risk.

Hypothesis Testing and Statistical Inference

Several statistical tests were conducted to determine whether observed relationships were statistically significant and not simply the result of random variation.

One-Sample t-Test

The one-sample t-test examined whether average stock returns differed significantly from zero.

For most selected assets, p-values exceeded the 0.05 significance level. Consequently, the null hypothesis could not be rejected for the majority of stocks.

This finding supports the weak form of market efficiency, suggesting that average daily returns are generally not significantly different from zero.

Paired t-Test

A paired t-test was performed to compare stock returns with BIST100 returns.

The null hypothesis stated that there is no significant difference between average stock returns and average market returns.

The results indicated that most selected stocks did not significantly outperform the market benchmark. This finding further supports the conclusion that market-wide dynamics dominate stock behavior within the Turkish equity market.

Pearson Correlation Test

Pearson correlation tests were used to evaluate the relationship between trading volume and volatility.

The results were statistically significant, allowing rejection of the null hypothesis.

This confirms that increased trading activity is associated with elevated volatility and market uncertainty.

Jarque-Bera Test

Jarque-Bera tests revealed strong evidence of non-normal return distributions.

The results confirmed:

- Heavy-tailed behavior
- Asymmetry
- Extreme observations

This finding is important because many traditional financial models rely on normality assumptions that may not hold in practice.

Augmented Dickey-Fuller Test

ADF tests confirmed that daily return series are stationary.

Stationarity is essential because non-stationary variables may produce misleading statistical relationships and unreliable predictive models.

The confirmation of stationarity improves the validity of subsequent regression and machine learning analyses.